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ManOxCo: From Oxygen Reliability Crisis to End-to-End Big Data Delivery

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Publication Date: 2026-04-19

Publication No.: MAHG-BD-PAPER-MOC-20260419-01

Abstract

This chapter presents the ManOxCo use case as the unifying business scenario for the complete Big Data course. The case is not framed as a generic technology exercise, but as a life-critical operational problem in which a liquid oxygen (LOX) dry-out at a hospital led to the death of a patient. From that event, the course builds a full end-to-end delivery journey in which students act as a simulated consultancy company selected to design and implement the ManOxCo data platform. The chapter examines the business problem, the likely chain of causation, the operational and governance failures behind the incident, and the rationale for organizing delivery through PI-1 planning and a twelve-lab execution path. Although the hands-on lab implementation stream is exercised through the ManTradeCoin training environment, every lab pattern is intentionally transferable to ManOxCo and is meant to be reinterpreted by students as part of a real customer-provider delivery relationship with defined outcomes, deliverables, and PI-1 commitments. The sequence begins with backlog-driven planning and platform foundations, continues through lakehouse engineering, streaming, governance, resilience, and analytical modeling, and culminates in an agentic AI assistant capable of consulting the platform in natural language over governed Gold-layer data products.

1. Introduction

ManOxCo is the central business case of the course because it forces students to think correctly about Big Data. The essential question is not which tool is fashionable, which cloud service is easiest to demo, or which architecture diagram appears most advanced. The

essential question is how to prevent a hospital from running out of medical oxygen. That change in emphasis matters. It moves the learning objective from technology admiration to operational responsibility.

In ManOxCo, data is not collected for its own sake. It exists to support the continuity of a life-critical supply chain that spans plants, trucks, hospital tanks, commercial contracts, maintenance windows, and eventually predictive and conversational intelligence. For that reason, the case provides a realistic structure for teaching end-to-end data platform design: business context, operational telemetry, batch and streaming ingestion, medallion architecture, dimensional modeling, governance, resilience, machine learning, and agentic AI.

As framed across the course, ManOxCo has already selected a consultancy company to execute the transformation. That consultancy is represented by the students. The pedagogical value of this framing is that it removes the ambiguity of a fictional sales pitch and replaces it with an execution mandate: the team has been chosen, PI-1 has been defined, the backlog exists, and delivery must proceed under architectural, governance, and time constraints. Even when the hands-on lab vehicle uses ManTradeCoin as the implementation sandbox, students are expected to interpret each exercise through the ManOxCo lens and treat the work as if they were delivering a professional platform solution to a real client under PI-1 scope.

2. The Business Problem

ManOxCo is an industrial gas company responsible for producing, storing, transporting, and delivering liquid oxygen to hospitals. Its operating model combines industrial production constraints with healthcare service obligations. Although ManOxCo manufactures and distributes oxygen as a commercial activity, its failures are judged by a much higher standard than those of an ordinary logistics network because its customers depend on uninterrupted oxygen availability for clinical care.

The triggering event behind the case is a hospital dry-out incident. A hospital supplied by ManOxCo ran out of available LOX, and a patient died. The event immediately transforms the meaning of the platform initiative. Forecasting, dashboards, and analytics are no longer abstract data outcomes; they become instruments of operational prevention, escalation, and accountability.

The case also establishes that the next platform solution must support an eighteen-month planning horizon while operating in real time. This combination is deliberate. ManOxCo needs long-range planning for plant capacity, storage, maintenance windows, and financial performance, but it also needs minute-level visibility into production and hospital consumption. A viable Big Data platform therefore has to unify strategic and operational decision-making rather than treating them as separate worlds.

3. Deep Dive Analysis of the Incident

3.1 Immediate Event Pattern

The incident is still formally under investigation, but the working hypothesis provided in the case is that an unauthorized maintenance shutdown at the Madrid LOX plant interrupted production and distribution. On its own, a maintenance event does not necessarily produce catastrophe. Industrial systems are expected to absorb maintenance through redundancy, coordination, and compensating actions. The critical issue is that the shutdown appears to have propagated through the rest of the operating chain without timely detection or successful mitigation.

3.2 Probable Root Cause Chain

The most useful way to analyze the event is not as a single-point failure but as a chain of reinforcing failures:

1. A plant-level operational change occurred without proper governance or synchronized planning.
2. Production capacity was reduced or interrupted at a key node in the supply network.
3. Distribution plans were not adjusted quickly enough across the remaining plants and trucks.
4. Hospital inventory depletion was not monitored at the level required to trigger timely intervention.
5. Decision-makers lacked a predictive mechanism that could estimate dry-out risk before the threshold was crossed.
6. Escalation failed to translate risk into action early enough to protect the affected hospital.

This sequence shows why ManOxCo is an excellent teaching case. The failure is not purely OT, purely IT, purely logistical, or purely analytical. It is systemic. A hospital dry-out emerges when governance, telemetry, operational planning, and decision intelligence fail to operate through one coordinated platform.

3.3 Systemic Root Causes

Several deeper causes can be inferred from the case documentation and the program baseline.

First, ManOxCo appears to have suffered from fragmented visibility. Plant production, storage levels, hospital consumption, delivery execution, and commercial master data belong to different systems and rhythms. Without a shared platform, each function sees only a partial truth.

Second, the company lacked an integrated IT/OT decision loop. OT data such as production telemetry and hospital consumption must be connected to IT data such as customer master records, pricing, costs, and planning assumptions. If those domains remain separated, operational warnings cannot be translated into business priorities and response actions.

Third, the organization lacked predictive and event-driven orchestration. A dry-out prevention system must do more than record history. It must detect abnormal depletion, project time-to-empty, consider truck and plant constraints, and recommend or trigger interventions before the situation becomes critical.

Fourth, maintenance governance appears weak. If a plant shutdown can occur without proper authorization, impact simulation, and contingency planning, then the company is vulnerable even before any data pipeline begins to run.

Fifth, the incident reveals a failure of business framing. The organization did not treat oxygen continuity as the dominant design principle. That omission is not merely strategic; it is architectural. When continuity of care is not the primary design criterion, systems tend to optimize for local efficiency instead of end-to-end risk prevention.

4. Business-Driven Before Data-Driven

The ManOxCo case is intentionally used to reinforce a foundational principle of the course:

“We must be business-driven before we become data-driven. In ManOxCo, the first design question is not which technology to deploy, but how to prevent a hospital dry-out and protect human life. Only when the mission is clear can architecture, data, and AI be aligned responsibly.”

Manuel Hernandez Giuliani

This statement matters because students often encounter Big Data through tooling. They learn Spark, storage layers, ingestion patterns, notebooks, dashboards, or AI frameworks, and then search for a problem to justify them. ManOxCo reverses that sequence. The life-critical business problem comes first. Technology is chosen afterward as a disciplined response.

That framing also explains why the final platform solution is a lakehouse and AI program rather than a single dashboard. Dry-out prevention requires historical analysis, real-time awareness, governed master data, analytical modeling, resilience, explainability, and human decision support. The architecture is therefore derived from the problem, not imposed on it.

5. Why the Course Uses a Simulated Consultancy Company

In the course narrative, students are not passive observers of the ManOxCo case. They operate as members of a simulated consultancy company that has already been selected to deliver the transformation. This framing has strong academic and professional value.

First, it mirrors how real enterprise work is executed. Once a consulting firm is selected, the work no longer revolves around abstract proposals. It revolves around backlog management, constraints, governance, milestones, architecture decisions, and visible delivery outcomes.

Second, it teaches accountability. A consultancy team cannot stop at describing the problem. It must structure the work, define responsibilities, justify trade-offs, and demonstrate progress through artifacts such as repositories, issues, project boards, architecture documents, and data products.

Third, it allows the course to begin from PI-1 planning rather than from a generic brainstorming phase. The baseline repository and PI-1 framing define the program objective, target architecture, workstreams, Definition of Done, expected outcomes, and delivery artifacts. Students inherit that mandate and must convert it into executable work.

The reference lab issue simulation reinforces this model. The lab sequence is organized as if the consultancy team were working through a real PI backlog: secure GitHub setup, architecture definition, environment build-out, ingestion, medallion engineering, Gold products, streaming, resilience, and AI. In practice, the hands-on implementation path is taught through the ManTradeCoin lab environment, but the consulting discipline, delivery logic, and architectural patterns are fully applicable to ManOxCo. This is exactly the kind of progression expected from a serious data consultancy operating under program increments rather than disconnected exercises.

6. PI-1 Planning as the Starting Point of Delivery

PI-1 planning is the conceptual beginning of the course platform solution, even if later lab files focus on the implementation stream. The reason is simple: ManOxCo cannot be solved by isolated technical tasks. It requires coordination across infrastructure, IT data, OT data, medallion design, governance, financial analysis, and AI.

Within the course, PI-1 planning establishes the delivery contract for the simulated consultancy:

- the mission is to prevent future LOX dry-outs
- the platform must unify IT and OT data
- the architecture must support batch and streaming
- the medallion model must lead to trustworthy Gold products
- the platform solution must include predictive and conversational intelligence
- delivery must be backlog-driven and reviewable

This is pedagogically important because students learn that architecture alone is insufficient. A platform becomes real only when planning, sequencing, roles, acceptance criteria, deliverables, and governance are explicit. The lab work therefore should not be understood as isolated notebook exercises. It should be understood as the execution body of a simulated consulting engagement in which each iteration produces artifacts that could be reviewed by a customer as part of PI-1.

7. The Twelve-Lab End-to-End Journey

The full course arc is best understood as a consulting program delivered in twelve labs. The hands-on lab implementation vehicle is ManTradeCoin, but that vehicle is used as a pedagogical scaffold rather than as the final business case. Each lab teaches a delivery pattern that students are expected to transfer to ManOxCo inside their simulated consultancy company, under a real-life customer-provider framing, with specific outcomes, deliverables, and PI-1 accountability.

Lab	Delivery Focus	ManOxCo Meaning
Lab 1	PI-1 planning and backlog framing	define mission, scope, workstreams, and execution model for the ManOxCo platform
Lab 2	GitHub workflow and project governance	establish the professional delivery operating model for the consulting team
Lab 3	Docker platform and local architecture	create the cloud-simulated technical foundation for the lakehouse platform solution
Lab 4	Spark fundamentals	build the engineering capability required to work with real ManOxCo data
Lab 5	Bronze ingestion	land raw ManOxCo source data with traceability and auditability
Lab 6	Silver engineering	clean, standardize, validate, and model trusted operational and business data
Lab 7	Gold dimensions	build reusable business context such as plants, hospitals, trucks, and dates
Lab 8	Gold facts and analytical serving	expose the core metrics and decision products needed by operations and leadership
Lab 9	Streaming ingestion	simulate the live OT behavior required for production telemetry and hospital consumption
Lab 10	Performance and resilience	ensure that the platform solution survives failures, latency, and operational stress
Lab 11	Local multi-agent RAG	introduce retrieval-grounded AI over ManOxCo knowledge and platform context
Lab 12	Gold-grounded agentic assistant	deliver a natural-language consulting experience over governed business data

This sequence is deliberately cumulative. Students do not jump directly to AI because the assistant would be hollow without a trustworthy data foundation. By the time the agentic chat experience is introduced, the consultancy team is expected to have produced the underlying

platform: raw data capture, curated datasets, analytical tables, and operational logic. In that sense, ManTradeCoin functions as the practical hands-on vehicle, while ManOxCo remains the executive and architectural business case against which students should justify relevance, transferability, and customer value. And to a final presentation to the ManOxCo Board with the outcome of their work.

8. From Lakehouse Engineering to Agentic AI

The final stage of the ManOxCo learning journey is especially important. The course does not end when students can ingest data or build dashboards. It ends when they can demonstrate that the entire platform solution supports executive and operational decision-making in natural language.

That is why the final outcome is an agentic AI chat grounded in the ManOxCo platform. The assistant is not supposed to invent answers from general language capability alone. It is expected to consult governed data products and domain knowledge, explain oxygen risk, support planning, and provide accountable recommendations.

In the ManOxCo setting, a mature natural-language interaction could include questions such as:

- Which hospitals are projected to fall below safe autonomy before the next planned delivery window?
- If the Madrid plant enters a maintenance window next month, which alternate supply paths create the lowest dry-out risk?
- Which delivery routes are operationally feasible but financially inefficient?
- What assumptions and source tables support the current risk recommendation?

These questions show why the course ends with agentic AI rather than with reporting alone. The goal is to demonstrate that the data platform has become consultable. In other words, the lakehouse is not merely storing information; it is enabling decision intelligence.

9. Educational and Architectural Significance

ManOxCo works as a course-wide case because it unifies academic rigor and professional realism. From an academic perspective, it allows students to connect concepts that are too often taught separately: distributed computing, medallion architecture, governance, IT/OT

integration, dimensional modeling, resilience engineering, predictive analytics, RAG, and agentic AI. From a professional perspective, it simulates what a real consultancy must do when confronted with a business-critical transformation program.

The course therefore teaches more than technical implementation. It teaches architectural responsibility. Students learn that a dry-out prevention platform cannot be judged only by code quality or tool choice. It must be judged by whether it shortens detection time, improves planning, supports escalation, reduces ambiguity, and ultimately protects human life.

10. Conclusion

The ManOxCo case is the backbone of the course because it transforms Big Data from an abstract technical domain into a disciplined response to a real operational crisis. A hospital dry-out is not a dashboard problem. It is a business, logistics, governance, and decision problem that requires an end-to-end platform.

By positioning students as a selected consultancy company working through PI-1 and then across twelve progressively integrated labs, the course creates a realistic delivery journey. It begins with planning and professional workflow, moves through lakehouse engineering and resilience, and ends with an agentic AI assistant capable of consulting the platform in natural language. The practical lab stream may be executed through ManTradeCoin, but the consulting discipline behind those labs is intended to remain fully reusable for ManOxCo as a real business case with customer expectations, deliverables, and acceptance boundaries.

What students build, therefore, is not just a sequence of labs. They build a complete argument about how data architecture can serve a life-critical mission. In ManOxCo, the ultimate measure of the platform is not technical elegance by itself, but whether the platform helps prevent the next dry-out before it can harm another patient.

Source Basis

This chapter is based on the ManOxCo baseline materials in this repository and the companion lab backlog framing referenced by the course:

- `slides/docs/ManOxCo/manoxco_use_case_en.md`
- `slides/docs/ManOxCo/manoxco_pi_1.md`
- `slides/docs/ManOxCo/data/data_source/README.md`
- `mahg_bigdata_training_lab/labs/blackboard/labs_issues.md`